Routing, capacity and width in dynamic convolution across five backbones

Cross-model summary of the replicated 132-run program

Ankit Kumar
Mechatronics Department, University of Agder, Grimstad, Norway

Generated from result files by `report/cross.py`

SYNTHETIC TEST DATA. Every number in this document is generated to test the pipeline. None is a result.

The 132-run ResNet-18 program (ten groups, 22 arms, six seeds) was repeated on each backbone with the same recipe, arms, seeds, margin and dynamic-layer rule. This summary puts 5 backbones side by side (652 runs, 965 GPU-hours logged). Every number is computed from the result files by the code that builds the per-model reports, so the two always agree. Backbones that did not pass the gate or have no results are listed in Table 1, not dropped.

1 The backbones

Table 1: Per backbone, CIFAR-100 width 64: stored parameters of the static and $K = 4$ `gap` networks, the share of the static network's parameters that sit in dynamic layers, the `gap` router's size relative to the static network, the gate run, the static mean over six seeds, runs present, diverged runs, groups that needed gradient accumulation, and logged GPU-hours.

backbone	static (M)	gap (M)	dyn. share (%)	router (% of static)	gate (%)	static mean (%)	runs	diverged	accum. groups	GPU-h
ResNet-18	11.22	44.46	98	2.6	78.05	78.34	132	0	0	204
MobileNetV2	2.35	3.68	3	48.4	73.78	73.92	130	1	0	190
VGG-16	14.77	59.29	100	2.7	73.01	73.04	130	1	0	190
ResNet-50	23.71	57.98	48	1.3	79.00	79.10	130	1	0	190
ConvNeXt-T	27.89	29.70	1	2.9	68.36	68.05	130	1	0	191

2 Bank against routing

The bank term is equivalent to zero (TOST, ± 0.5 pp) in 24 of 25 backbone–condition cells, and the routing term is positive with $p < 0.05$ in 24 of 25 (Tables 2, 3).

Table 2: Bank term (pp); $\equiv$ marks equivalence to zero by TOST.

backbone	C16	C32	C64	T32	T64
ResNet-18	-0.223 $\equiv$	+0.175 $\equiv$	+0.048 $\equiv$	+0.107 $\equiv$	+0.053 $\equiv$
MobileNetV2	+0.073 $\equiv$	-0.159 $\equiv$	+0.156 $\equiv$	+0.121 $\equiv$	+0.099 $\equiv$
VGG-16	+0.128 $\equiv$	+0.240 $\equiv$	-0.043 $\equiv$	-0.010 $\equiv$	+0.136 $\equiv$
ResNet-50	+0.259 $\equiv$	-0.018 $\equiv$	+0.007 $\equiv$	+0.118 $\equiv$	+0.042 $\equiv$
ConvNeXt-T	-0.007 $\equiv$	+0.073 $\equiv$	+0.066 $\equiv$	+0.060 $\equiv$	+0.127

Table 3: Routing term (pp); * marks $p < 0.05$.

backbone	C16	C32	C64	T32	T64
ResNet-18	+0.755 *	+0.978 *	+0.032	+2.078 *	+2.228 *
MobileNetV2	+0.863 *	+0.937 *	+0.659 *	+0.830 *	+0.564 *
VGG-16	+0.933 *	+0.803 *	+1.027 *	+0.807 *	+0.819 *
ResNet-50	+0.694 *	+0.973 *	+0.737 *	+0.875 *	+0.789 *
ConvNeXt-T	+0.974 *	+1.015 *	+0.669 *	+1.011 *	+0.840 *

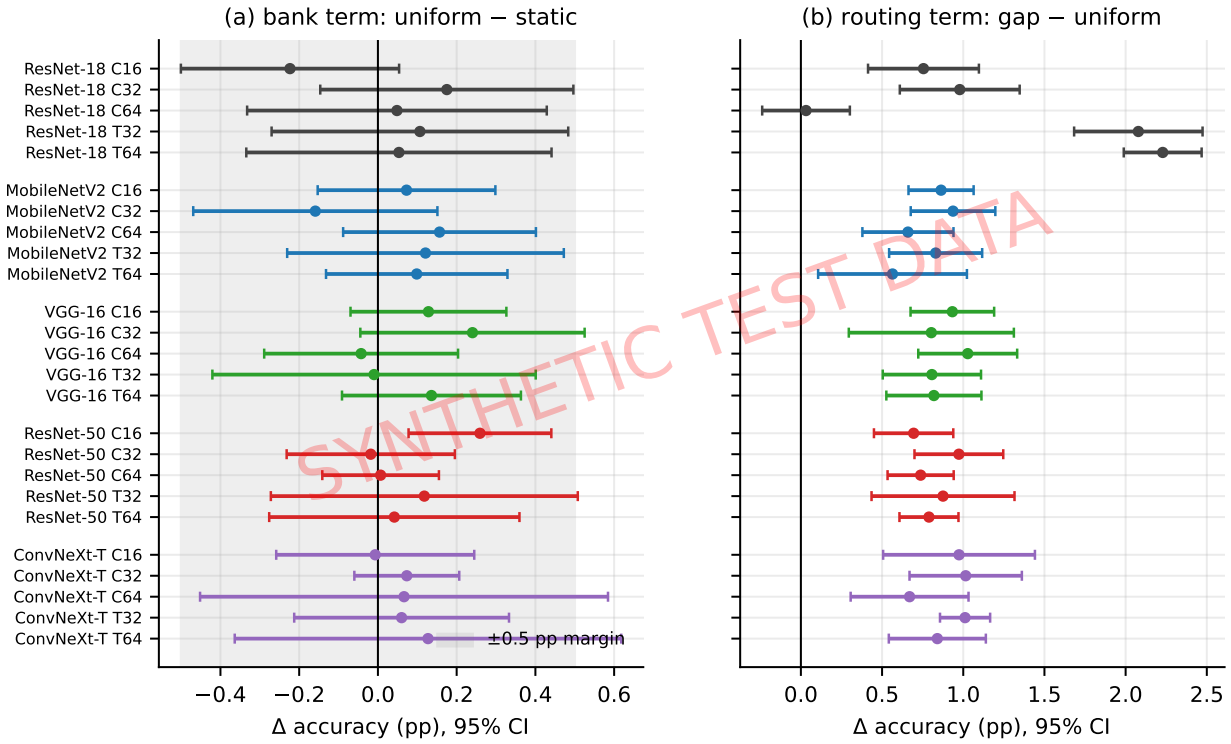


Figure 1: (a) Bank term, `uniform−static`, and (b) routing term, `gap−uniform`, for every backbone and condition (C: CIFAR-100, T: Tiny-ImageNet, number: width), 95 % confidence intervals.

3 Total effect by width

Table 4: `gap−static` (pp) and verdict per condition.

backbone	C16	C32	C64	T32	T64
ResNet-18	+0.532 (higher)	+1.153 (higher)	+0.080 (equivalent)	+2.185 (higher)	+2.282 (higher)
MobileNetV2	+0.936 (higher)	+0.778 (higher)	+0.816 (higher)	+0.951 (higher)	+0.744 (higher)
VGG-16	+1.061 (higher)	+1.017 (higher)	+0.985 (higher)	+0.797 (higher)	+0.955 (higher)
ResNet-50	+0.954 (higher)	+0.955 (higher)	+0.744 (higher)	+1.044 (higher (n=5))	+0.830 (higher)
ConvNeXt-T	+0.967 (higher)	+1.104 (higher (n=5))	+0.736 (higher)	+1.071 (higher)	+0.953 (higher (n=5))

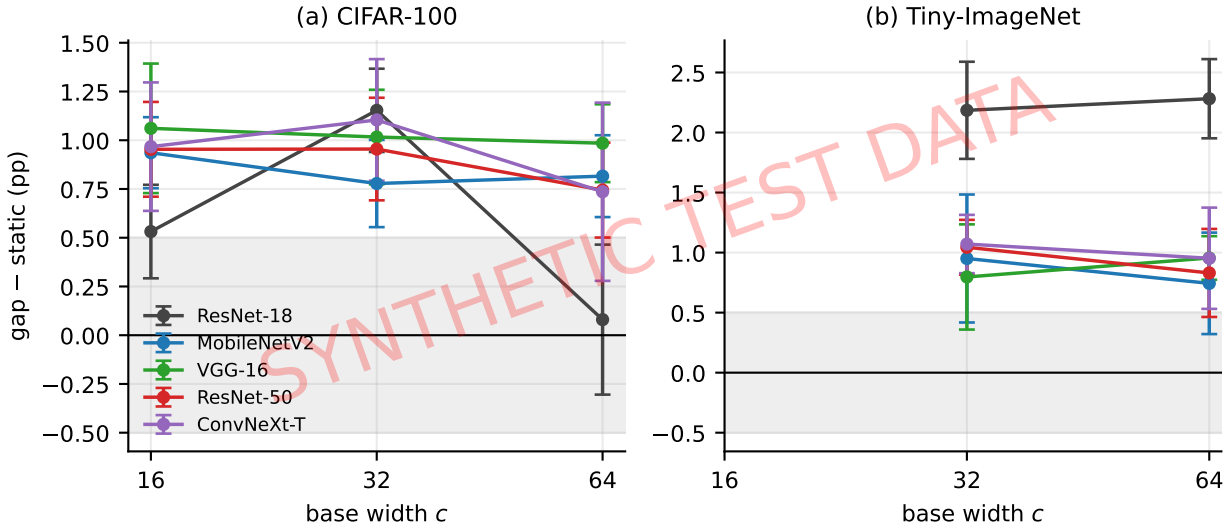


Figure 2: `gap−static` against width, every backbone, 95 % CI; the band is the margin.

4 Equal parameter budgets

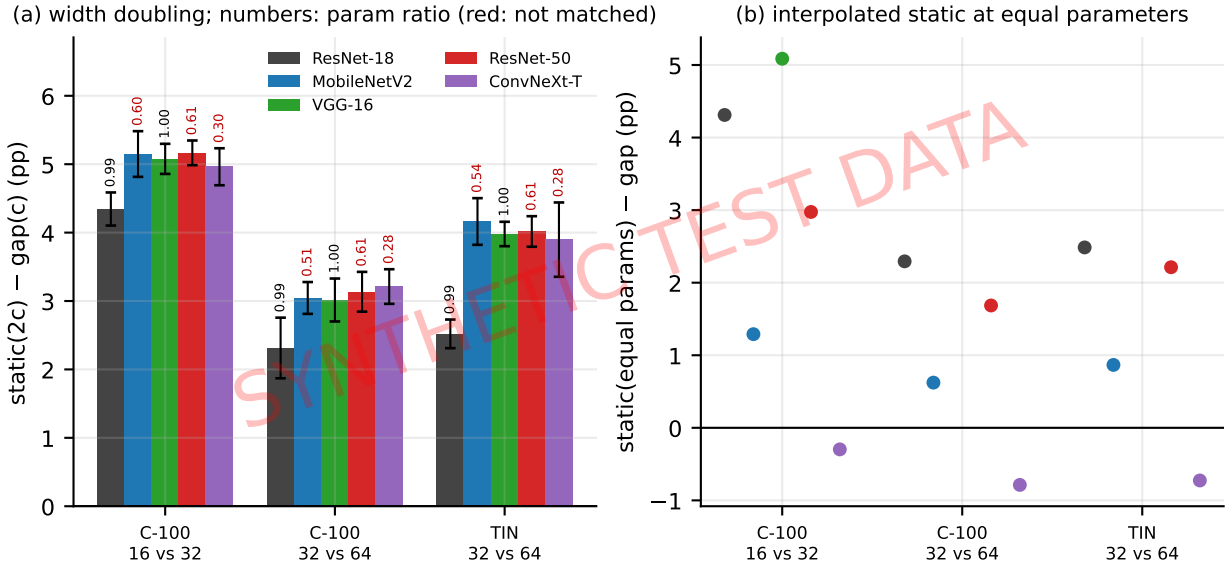


Figure 3: (a) `static` at width $2c$ minus `gap` at width c, 95 % CI; the number under each bar is the parameter ratio routed/plain, red where it is outside ± 10 %. (b) The static accuracy curve interpolated in log-parameters at the `gap` network's count, minus `gap`; missing points fall outside the static range.

Width doubling is a matched-budget comparison only where nearly all parameters sit in dynamic layers. It is matched (within 10 %) in 6 of 15 backbone–pair cells (Table 5); elsewhere the table gives the ratio and the interpolated equal-parameter comparison.

Table 5: `static`$(2c)$ − `gap`(c), parameter ratio and the interpolated comparison at equal parameters (no CI).

backbone	pair	Δ (pp)	95 % CI	p	ratio	label	Δ at equal params
ResNet-18	CIFAR-100 w16/w32	+4.345	[+4.103, +4.587]	$< 10^{-4}$	0.99	matched	+4.311
ResNet-18	CIFAR-100 w32/w64	+2.313	[+1.870, +2.757]	$< 10^{-4}$	0.99	matched	+2.294
ResNet-18	Tiny-ImageNet w32/w64	+2.520	[+2.311, +2.729]	$< 10^{-4}$	0.99	matched	+2.486
MobileNetV2	CIFAR-100 w16/w32	+5.149	[+4.815, +5.483]	$< 10^{-4}$	0.60	not matched	+1.291
MobileNetV2	CIFAR-100 w32/w64	+3.045	[+2.812, +3.277]	$< 10^{-4}$	0.51	not matched	+0.623
MobileNetV2	Tiny-ImageNet w32/w64	+4.163	[+3.822, +4.504]	$< 10^{-4}$	0.54	not matched	+0.867
VGG-16	CIFAR-100 w16/w32	+5.078	[+4.858, +5.298]	$< 10^{-4}$	1.00	matched	+5.087
VGG-16	CIFAR-100 w32/w64	+3.015	[+2.701, +3.329]	$< 10^{-4}$	1.00	matched	–
VGG-16	Tiny-ImageNet w32/w64	+3.980	[+3.802, +4.158]	$< 10^{-4}$	1.00	matched	–
ResNet-50	CIFAR-100 w16/w32	+5.166	[+4.986, +5.347]	$< 10^{-4}$	0.61	not matched	+2.975
ResNet-50	CIFAR-100 w32/w64	+3.135	[+2.844, +3.426]	$< 10^{-4}$	0.61	not matched	+1.686
ResNet-50	Tiny-ImageNet w32/w64	+4.017	[+3.793, +4.240]	$< 10^{-4}$	0.61	not matched	+2.213
ConvNeXt-T	CIFAR-100 w16/w32	+4.963	[+4.692, +5.233]	$< 10^{-4}$	0.30	not matched	-0.297
ConvNeXt-T	CIFAR-100 w32/w64	+3.212	[+2.959, +3.465]	$< 10^{-4}$	0.28	not matched	-0.787
ConvNeXt-T	Tiny-ImageNet w32/w64	+3.897	[+3.354, +4.441]	$< 10^{-4}$	0.28	not matched	-0.726

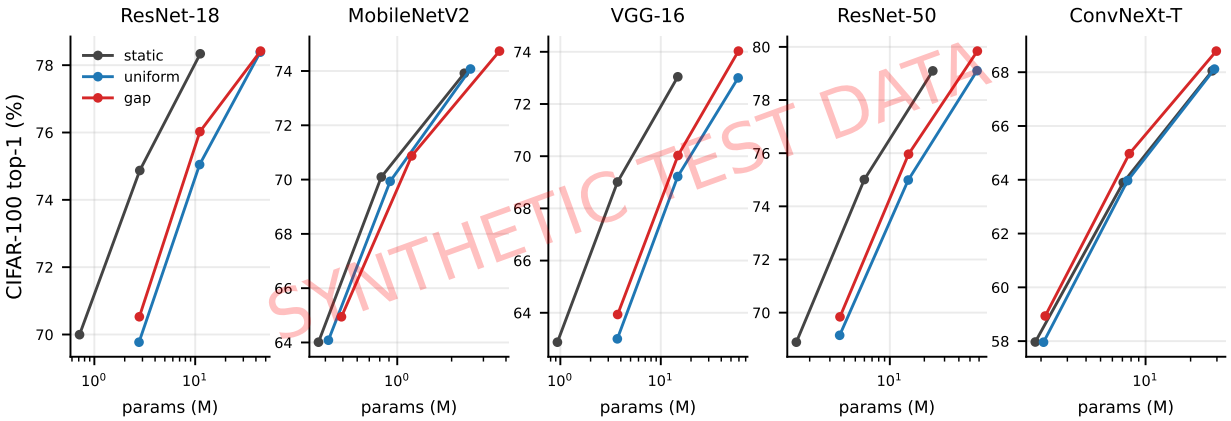


Figure 4: CIFAR-100 accuracy against stored parameters, widths 16, 32 and 64, per backbone.

5 Temperature, bank size and routing entropy

Table 6: Annealed − constant (pp, p) per width, and $K = 8 - K = 2$.

backbone	w16	w32	w64	$K = 8 - K = 2$	verdict
ResNet-18	+0.115 (0.4313)	-0.682 (0.0017)	-0.048 (0.7449)	+0.098	equivalent
MobileNetV2	-0.509 (0.0110)	-0.314 (0.0723)	-0.179 (0.1592)	+0.167	equivalent
VGG-16	-0.269 (0.0246)	-0.371 (0.0196)	-0.525 (0.0061)	+0.144	higher (n=5)
ResNet-50	-0.307 (0.0448)	-0.358 (0.0585)	-0.152 (0.0711)	-0.100	equivalent
ConvNeXt-T	-0.463 (0.0012)	-0.558 (0.0059)	-0.050 (0.6872)	+0.144	equivalent

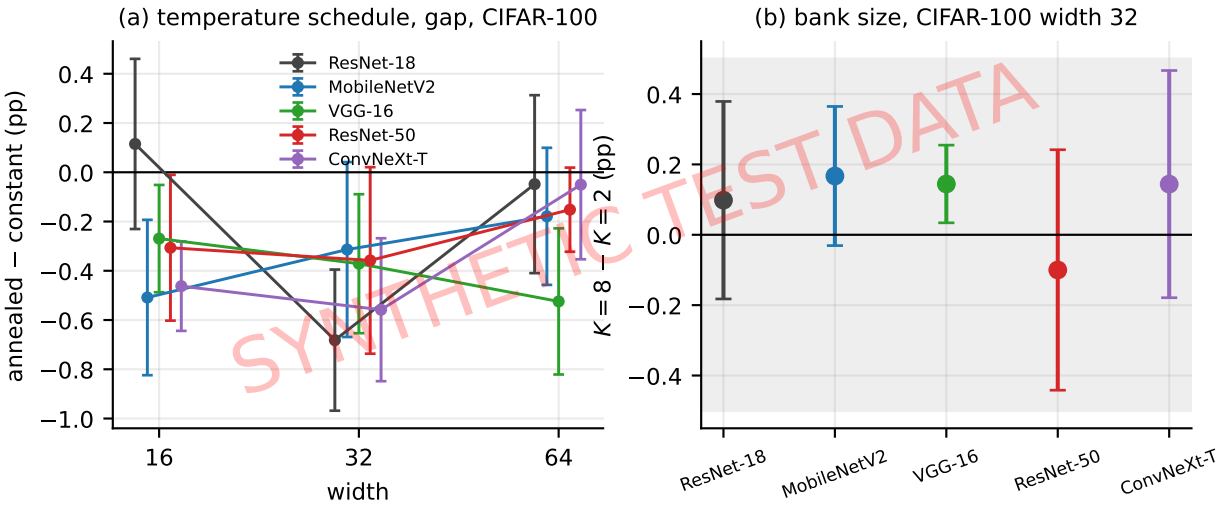


Figure 5: (a) Annealed minus constant temperature, `gap`, CIFAR-100. (b) $K = 8$ minus $K = 2$, CIFAR-100 width 32.

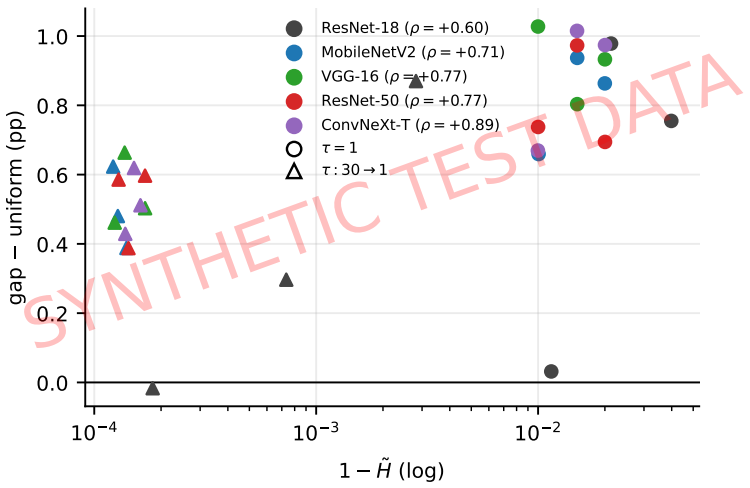


Figure 6: Router flatness $1 - \tilde{H}$ against the benefit over `uniform`, CIFAR-100, three widths and two recipes per backbone; ρ is Spearman's correlation within a backbone.

Table 7: Entropy against benefit: points, Spearman ρ, and at width 16 the ratio of constant to annealed $1 - \tilde{H}$.

backbone	points	ρ	flatness ratio (w16)
ResNet-18	6	+0.60	14.1
MobileNetV2	6	+0.71	143.1
VGG-16	6	+0.77	145.9
ResNet-50	6	+0.77	140.4
ConvNeXt-T	6	+0.89	124.0

6 Pre-registered predictions

Recorded in `PREREGISTRATION_MODELS.md` on 30 September 2026 for the 4 new backbones before any of their runs. P1: the bank term is equivalent to zero in all five conditions (held for 3 of 4). P2: `static` at width $2c$ beats `gap` at width c in all three pairs (held for 4). P3: the routing term is positive with $p < 0.05$ in at least three conditions (held for 4). ResNet-18 is shown for reference; for it these are post hoc.

Table 8: Pre-registered predictions per backbone.

	P1 bank $\equiv 0$		P2 width wins		P3 routing > 0	
backbone	count	outcome	count	outcome	count	outcome
ResNet-18 (post hoc)	5 of 5	held	3 of 3	held	4 of 5	held
MobileNetV2	5 of 5	held	3 of 3	held	5 of 5	held
VGG-16	5 of 5	held	3 of 3	held	5 of 5	held
ResNet-50	5 of 5	held	3 of 3	held	5 of 5	held
ConvNeXt-T	4 of 5	failed	3 of 3	held	5 of 5	held

7 Threats to validity

One recipe for five architectures. Every backbone is trained with the ResNet-18 recipe (SGD, lr 0.1, 150 epochs). That is the design: a per-architecture recipe would change two things at once. It also means absolute accuracies, above all ConvNeXt-Tiny's, sit below what a tuned recipe reaches. The gate guards against a backbone the recipe does not train at all; it does not make the recipes equally good.

Placement. The ResNet-18 rule makes different shares of each network dynamic (Table 1): nearly all of ResNet-18 and VGG-16, a smaller part of ResNet-50, and only depthwise layers in MobileNetV2 and ConvNeXt-Tiny. A backbone-to-backbone difference may come from this share rather than from the architecture.

Router size. Every backbone uses the ResNet-18 router, a two-layer MLP of hidden width $C_{\text{in}}/4$. Its size grows with the square of the input channels while a depthwise kernel grows linearly, so in the depthwise backbones the router is much larger than the bank it routes (Table 1). The routing term `gap−uniform` then also compares networks of very different stored size; `frozen_random` (an untrained router, held as buffers) is the control for this at CIFAR-100 width 64.

Initialisation. In the 132 ResNet-18 runs static convolutions kept PyTorch's default initialisation while bank kernels were Kaiming-normal. The new backbones use one scheme for both. The routing term (`gap−uniform`) is unaffected in either case, since both arms have banks; the bank term of ResNet-18 compares two initialisations as well as two architectures.

Gradient accumulation. Where batch 128 did not fit, a group ran with micro-batches and accumulation (Table 1). The optimiser steps and schedule are unchanged; BatchNorm statistics are per micro-batch. The choice is per group, so every paired comparison within a group shares it.